

Conditional Probability

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First, an example

Roll a fair six-sided die. Let $A = \{\omega : \omega \text{ is even}\}$ and $B = \{\omega : \omega \geq 4\}$. Then $A \cap B = \{4, 6\}$, so

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3}.$$

Read aloud: “probability of A given B equals probability of A intersect B divided by probability of B, which is two-sixths over three-sixths, equals two-thirds.” Knowing that B occurred shifts our belief about A from $1/2$ (the unconditional probability of an even roll) to $2/3$. Conditional probability is exactly this update rule, and the rest of the lesson develops it carefully.

1 Setup

Throughout, let $(\Omega, \mathcal{F}, P)$ be a probability space. We use $A, B, \dots$ for elements of $\mathcal{F}$ (“events”).

2 Definitions

Definition 1 (Conditional probability). Let $B \in \mathcal{F}$ with $P(B) > 0$. For any $A \in \mathcal{F}$, the *conditional probability of A given B* is

$$P(A | B) := \frac{P(A \cap B)}{P(B)}.$$

Read aloud: “probability of A given B is defined as probability of A intersect B, divided by probability of B.”

Proposition 2 ($P(\cdot | B)$ is a probability measure). Fix B with $P(B) > 0$. The set function $Q : \mathcal{F} \rightarrow [0, 1]$ given by $Q(A) = P(A | B)$ is a probability measure on $(\Omega, \mathcal{F})$.

Proof. Non-negativity: $P(A \cap B) \geq 0$ and $P(B) > 0$, so $Q(A) \geq 0$. Normalization: $Q(\Omega) = P(\Omega \cap B)/P(B) = P(B)/P(B) = 1$. Countable additivity: if $\{A_n\}$ are pairwise disjoint, then so are $\{A_n \cap B\}$, so

$$Q\left(\bigcup_n A_n\right) = \frac{P(\bigcup_n (A_n \cap B))}{P(B)} = \frac{\sum_n P(A_n \cap B)}{P(B)} = \sum_n Q(A_n). \quad \square$$

Read aloud: “Q of the union over n of A-sub-n equals P of the union over n of A-sub-n intersect B, all over P of B; which equals the sum over n of P of A-sub-n intersect B, over P of B; which equals the sum over n of Q of A-sub-n.”

Definition 3 (Independence). Events A, B are *independent* iff $P(A \cap B) = P(A)P(B)$. Equivalently, when $P(B) > 0$, iff $P(A | B) = P(A)$.

3 Total Probability and Bayes

Lemma 4 (Law of total probability). *Let $\{B_i\}_{i=1}^n$ be a finite measurable partition of Ω with $P(B_i) > 0$ for all i . For any $A \in \mathcal{F}$,*

$$P(A) = \sum_{i=1}^n P(A \mid B_i) P(B_i).$$

Read aloud: “probability of A equals the sum from i equals one to n of probability of A given B-sub-i, times probability of B-sub-i.”

Proof. Since $\{B_i\}$ partitions Ω , the sets $\{A \cap B_i\}$ partition A . By finite additivity of P and the definition of conditional probability,

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i) P(B_i). \quad \square$$

Read aloud: “probability of A equals the sum over i of probability of A intersect B-sub-i, which equals the sum over i of probability of A given B-sub-i, times probability of B-sub-i.”

Theorem 5 (Bayes). *Let $A, B \in \mathcal{F}$ with $P(A) > 0$ and $P(B) > 0$. Then*

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}.$$

Read aloud: “probability of A given B equals probability of B given A, times probability of A, all over probability of B.” Moreover, if $\{A_i\}_{i=1}^n$ is a measurable partition of Ω with each $P(A_i) > 0$, then for any j ,

$$P(A_j \mid B) = \frac{P(B \mid A_j) P(A_j)}{\sum_{i=1}^n P(B \mid A_i) P(A_i)}.$$

Read aloud: “probability of A-sub-j given B equals probability of B given A-sub-j, times probability of A-sub-j, all over the sum from i equals one to n of probability of B given A-sub-i, times probability of A-sub-i.”

Proof. By definition,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

Read aloud: “probability of A given B equals probability of A intersect B, over probability of B; and probability of B given A equals probability of A intersect B, over probability of A.” The second identity yields $P(A \cap B) = P(B \mid A) P(A)$. Substituting into the first gives

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)},$$

Read aloud: “probability of A given B equals probability of B given A, times probability of A, all over probability of B.” which is the first claim. For the second, apply the law of total probability with the partition $\{A_i\}$ to expand $P(B) = \sum_i P(B \mid A_i) P(A_i)$ in the denominator. $\square$

4 Chain rule

Proposition 6 (Multiplication / chain rule). *Let $A_1, \dots, A_n \in \mathcal{F}$ with $P(A_1 \cap \dots \cap A_{n-1}) > 0$. Then*

$$P(A_1 \cap \dots \cap A_n) = P(A_1) \prod_{k=2}^n P(A_k \mid A_1 \cap \dots \cap A_{k-1}).$$

Read aloud: “probability of A-one intersect dot dot dot intersect A-n equals probability of A-one, times the product from k equals two to n of probability of A-sub-k given A-one intersect dot dot dot intersect A-sub-k-minus-one.”

Proof. Induct on n . The base $n = 2$ is the definition rearranged: $P(A_1 \cap A_2) = P(A_2 \mid A_1)P(A_1)$. For the inductive step, apply the base case to $A_1 \cap \dots \cap A_{n-1}$ and A_n . $\square$

5 Example

Example 7 (Monty Hall). Three doors; a car uniformly hidden behind one. The contestant picks door 1; the host (knowing the layout) opens a goat door from $\{2, 3\}$, breaking ties uniformly. Conditional on the host opening door 3, Bayes gives $P(\text{car at 2} \mid \text{open 3}) = 2/3$. Switching is optimal.